

Bagging — Short Notes

What is Bagging?

Bagging = Bootstrap Aggregating

It is an **ensemble technique** that trains multiple models on different subsets of the training data and combines their predictions.

How does Bagging work?

Step 1 — Bootstrapping

- Create multiple datasets from the original dataset.
- Sample **with replacement**.
- Each dataset is used to train a separate model.

Step 2 — Train Models

- Train the same type of base model independently on each sample.

Step 3 — Aggregation

- **Classification → Majority voting**
- **Regression → Average**

Example

Original data:

$D = [1, 2, 3, 4, 5]$

Bootstrap samples could be:

$D1 = [1, 2, 2, 4, 5]$

$D2 = [2, 3, 3, 5, 1]$

$D3 = [4, 4, 1, 2, 5]$

Train a separate Decision Tree on each → combine their predictions.

Why Bagging?

Bagging is especially useful for **high-variance models** such as Decision Trees.

It:

-  Reduces variance
-  Reduces overfitting
-  Improves stability
-  Usually improves prediction performance

{"learning_viz":{"type_id":"VARIANCE"}}

Types of Bagging

Technique	What is sampled?
Bagging	Rows with replacement
Pasting	Rows without replacement
Random Subspaces	Features/columns
Random Patches	Rows + features

★ Easy Memory

Bagging = Same model + different data samples + parallel training + aggregate predictions

🌲 **Random Forest is a famous example of Bagging**, using multiple Decision Trees with additional feature randomness.